AIMO Proof Pilot — Final Grade

Problem (#_ID): 3_SINDEX **Submission:** model_4

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{0}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

The solution introduces a reasonable recurrence for the partition multiplicities, but the proof of $N \leq 5$ is invalid: the step claiming that equations (4)–(5) force all multiplicities $m(v)$ to be equal does not follow, and this collapses the entire upper-bound argument. The accompanying construction is also incorrect — it produces negative multiplicities (the write-up itself asserts $0 - 1 = 0$) and fails, for instance at $i = 1$. With neither a valid upper bound nor a valid construction, there is nothing to credit, and the submission falls short of the partial-progress milestone. Scores **0**.